

Powers and roots of complex numbers — Solutions

Warm-up

Question 1:

De Moivre's Theorem states that for any complex number $z = r(\cos \theta + i \sin \theta)$ and any integer n , the n -th power of z is given by:

$$\operatorname{cis}(n\theta) = (\cos \theta + i \sin \theta)^n$$

Or more generally, if $z = r\operatorname{cis}(\theta)$, then $z^n = r^n\operatorname{cis}(n\theta)$. This allows us to relate the power of a complex number to the multiplication of its argument and the exponentiation of its modulus.

Question 2a:

Given $z^5 = 32\operatorname{cis}(\frac{\pi}{3})$. To find one solution, we apply De Moivre's theorem in reverse (taking the 5th root):

$$\begin{aligned} z &= (32\operatorname{cis}(\frac{\pi}{3}))^{1/5} \\ &= 32^{1/5}\operatorname{cis}(\frac{\pi/3}{5}) \\ &= 2\operatorname{cis}(\frac{\pi}{15}) \end{aligned}$$

Question 2b:

Let $z_1 = 2\operatorname{cis}(\frac{\pi}{15})$. Rotating z_1 anticlockwise by $\frac{2\pi}{5}$ is equivalent to multiplying by $\operatorname{cis}(\frac{2\pi}{5})$. Let the new solution be z_2 :

$$\begin{aligned} z_2 &= 2\operatorname{cis}(\frac{\pi}{15}) \cdot \operatorname{cis}(\frac{2\pi}{5}) \\ &= 2\operatorname{cis}(\frac{\pi}{15} + \frac{6\pi}{15}) \\ &= 2\operatorname{cis}(\frac{7\pi}{15}) \end{aligned}$$

Now we check if $z_2^5 = 32\operatorname{cis}(\frac{\pi}{3})$:

$$\begin{aligned} z_2^5 &= (2\operatorname{cis}(\frac{7\pi}{15}))^5 \\ &= 2^5\operatorname{cis}(5 \cdot \frac{7\pi}{15}) \\ &= 32\operatorname{cis}(\frac{7\pi}{3}) \\ &= 32\operatorname{cis}(\frac{7\pi}{3} - 2\pi) \\ &= 32\operatorname{cis}(\frac{\pi}{3}) \end{aligned}$$

Thus, z_2 also satisfies the equation.

Question 2c:

The solutions are found by repeatedly adding $\frac{2\pi}{5}$ to the argument of the first solution $z_1 = 2\text{cis}(\frac{\pi}{15})$:

$$\begin{aligned} z_1 &= 2\text{cis}\left(\frac{\pi}{15}\right) \\ z_2 &= 2\text{cis}\left(\frac{\pi}{15} + \frac{6\pi}{15}\right) = 2\text{cis}\left(\frac{7\pi}{15}\right) \\ z_3 &= 2\text{cis}\left(\frac{7\pi}{15} + \frac{6\pi}{15}\right) = 2\text{cis}\left(\frac{13\pi}{15}\right) \\ z_4 &= 2\text{cis}\left(\frac{13\pi}{15} + \frac{6\pi}{15}\right) = 2\text{cis}\left(\frac{19\pi}{15}\right) \\ z_5 &= 2\text{cis}\left(\frac{19\pi}{15} + \frac{6\pi}{15}\right) = 2\text{cis}\left(\frac{25\pi}{15}\right) = 2\text{cis}\left(\frac{5\pi}{3}\right) \end{aligned}$$

Question 2d:

Rotating by $\frac{2\pi}{5}$ produces new solutions because the n -th roots of a complex number are equally spaced around a circle in the complex plane, separated by $\frac{2\pi}{n}$ radians. When we raise the rotated number to the 5th power, the rotation angle $\frac{2\pi}{5}$ is multiplied by 5, resulting in a full 2π rotation, which returns the number to the original position on the unit circle.

If we produce a sixth solution by rotating further:

$$\begin{aligned} z_6 &= 2\text{cis}\left(\frac{25\pi}{15} + \frac{6\pi}{15}\right) = 2\text{cis}\left(\frac{31\pi}{15}\right) \\ &= 2\text{cis}\left(\frac{31\pi}{15} - 2\pi\right) = 2\text{cis}\left(\frac{\pi}{15}\right) = z_1 \end{aligned}$$

We return to the first solution, meaning there are only 5 unique solutions.

Question 2e:

To ensure all solutions have arguments between $-\pi$ and π , we can subtract 2π from any argument θ where $\pi < \theta \leq 2\pi$.

From part (c), the arguments are $\frac{\pi}{15}, \frac{7\pi}{15}, \frac{13\pi}{15}, \frac{19\pi}{15}, \frac{25\pi}{15}$.

$$\begin{aligned} \frac{19\pi}{15} &\implies \frac{19\pi}{15} - 2\pi = \frac{19\pi - 30\pi}{15} = -\frac{11\pi}{15} \\ \frac{25\pi}{15} &\implies \frac{25\pi}{15} - 2\pi = \frac{25\pi - 30\pi}{15} = -\frac{5\pi}{15} = -\frac{\pi}{3} \end{aligned}$$

Thus, the solutions in principal argument form are $2\text{cis}(\frac{\pi}{15}), 2\text{cis}(\frac{7\pi}{15}), 2\text{cis}(\frac{13\pi}{15}), 2\text{cis}(-\frac{11\pi}{15}), 2\text{cis}(-\frac{\pi}{3})$.

Question 3:

We seek the 5th roots of $z = 8\text{cis}(\frac{5\pi}{4})$. The general form of the roots is $z_k = r^{1/n}\text{cis}(\frac{\theta+2k\pi}{n})$ for $k = 0, 1, 2, 3, 4$.

$$\text{Modulus: } r^{1/5} = 8^{1/5}$$

$$\text{Arguments: } \theta_k = \frac{\frac{5\pi}{4} + 2k\pi}{5} = \frac{5\pi + 8k\pi}{20}$$

Calculating for each k :

$$\begin{aligned}
 k = 0 : z_0 &= 8^{1/5} \operatorname{cis}\left(\frac{5\pi}{20}\right) = 8^{1/5} \operatorname{cis}\left(\frac{\pi}{4}\right) \\
 k = 1 : z_1 &= 8^{1/5} \operatorname{cis}\left(\frac{13\pi}{20}\right) \\
 k = 2 : z_2 &= 8^{1/5} \operatorname{cis}\left(\frac{21\pi}{20}\right) = 8^{1/5} \operatorname{cis}\left(-\frac{19\pi}{20}\right) \\
 k = 3 : z_3 &= 8^{1/5} \operatorname{cis}\left(\frac{29\pi}{20}\right) = 8^{1/5} \operatorname{cis}\left(-\frac{11\pi}{20}\right) \\
 k = 4 : z_4 &= 8^{1/5} \operatorname{cis}\left(\frac{37\pi}{20}\right) = 8^{1/5} \operatorname{cis}\left(-\frac{3\pi}{20}\right)
 \end{aligned}$$

Question 4:

We seek the cube roots of -1 . First, express -1 in modulus-argument form:

$$-1 = 1 \operatorname{cis}(\pi)$$

Using the root formula $z_k = r^{1/n} \operatorname{cis}\left(\frac{\theta + 2k\pi}{n}\right)$ for $k = 0, 1, 2$:

$$\begin{aligned}
 z_0 &= 1^{1/3} \operatorname{cis}\left(\frac{\pi}{3}\right) = \cos\left(\frac{\pi}{3}\right) + i \sin\left(\frac{\pi}{3}\right) = \frac{1}{2} + i \frac{\sqrt{3}}{2} \\
 z_1 &= 1^{1/3} \operatorname{cis}\left(\frac{\pi + 2\pi}{3}\right) = \operatorname{cis}(\pi) = -1 + 0i \\
 z_2 &= 1^{1/3} \operatorname{cis}\left(\frac{\pi + 4\pi}{3}\right) = \operatorname{cis}\left(\frac{5\pi}{3}\right) = \cos\left(\frac{5\pi}{3}\right) + i \sin\left(\frac{5\pi}{3}\right) = \frac{1}{2} - i \frac{\sqrt{3}}{2}
 \end{aligned}$$

Question 5a:

Given $z^7 = 1$. Express 1 as $1 \operatorname{cis}(2\pi)$:

$$\begin{aligned}
 z &= (1 \operatorname{cis}(2\pi))^{1/7} \\
 &= 1^{1/7} \operatorname{cis}\left(\frac{2\pi}{7}\right) \\
 &= \operatorname{cis}\left(\frac{2\pi}{7}\right)
 \end{aligned}$$

This is a non-real solution since $\sin\left(\frac{2\pi}{7}\right) \neq 0$.

Question 5b:

Let $\omega = \operatorname{cis}\left(\frac{2\pi}{7}\right)$. We want to show ω^2 satisfies $z^7 = 1$:

$$\begin{aligned}
 (\operatorname{cis}\left(\frac{2\pi}{7}\right))^{2 \cdot 7} &= (\operatorname{cis}\left(\frac{2\pi}{7}\right))^{14} \\
 &= \operatorname{cis}\left(\frac{2\pi \cdot 14}{7}\right) \\
 &= \operatorname{cis}(4\pi) \\
 &= 1
 \end{aligned}$$

Thus, ω^2 is a solution.

Question 5c:

The 7 solutions are $1, \omega, \omega^2, \omega^3, \omega^4, \omega^5, \omega^6$ where $\omega = \text{cis}(\frac{2\pi}{7})$.

Plotting these on an Argand diagram, they all have modulus 1 and are separated by $\frac{2\pi}{7}$ radians. They form the vertices of a regular heptagon inscribed in the unit circle.

Question 5d:

The sum of the n -th roots of unity is always 0. This can be shown using the geometric series formula:

$$\begin{aligned}\text{Sum} &= 1 + \omega + \omega^2 + \omega^3 + \omega^4 + \omega^5 + \omega^6 \\ &= \frac{1(\omega^7 - 1)}{\omega - 1} \\ &= \frac{1 - 1}{\omega - 1} \\ &= 0\end{aligned}$$

Question 6:

Given $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$. We want to simplify $(1 - \omega^8)(1 - \omega^4)(1 - \omega^2)(1 - \omega)$.

First, simplify the powers of ω using $\omega^3 = 1$:

$$\begin{aligned}\omega^8 &= (\omega^3)^2 \cdot \omega^2 = 1^2 \cdot \omega^2 = \omega^2 \\ \omega^4 &= \omega^3 \cdot \omega = 1 \cdot \omega = \omega\end{aligned}$$

Substitute these back into the expression:

$$\begin{aligned}\text{Expr} &= (1 - \omega^2)(1 - \omega)(1 - \omega^2)(1 - \omega) \\ &= [(1 - \omega)(1 - \omega^2)]^2 \\ &= [1 - \omega^2 - \omega + \omega^3]^2 \\ &= [1 - (\omega + \omega^2) + 1]^2 \\ &= [2 - (-1)]^2 \\ &= [3]^2 = 9\end{aligned}$$

Question 7:

Given $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$. Simplify $(1 + 2\omega - 3\omega^2)(1 + 3\omega - 2\omega^2)$.

Use the identity $\omega^2 = -1 - \omega$ to eliminate ω^2 from both brackets:

$$\begin{aligned}\text{Bracket 1: } 1 + 2\omega - 3\omega^2 &= 1 + 2\omega - 3(-1 - \omega) \\ &= 1 + 2\omega + 3 + 3\omega \\ &= 4 + 5\omega \\ \text{Bracket 2: } 1 + 3\omega - 2\omega^2 &= 1 + 3\omega - 2(-1 - \omega) \\ &= 1 + 3\omega + 2 + 2\omega \\ &= 3 + 5\omega\end{aligned}$$

Now multiply the simplified brackets:

$$\begin{aligned}
 \text{Expr} &= (4 + 5\omega)(3 + 5\omega) \\
 &= 12 + 20\omega + 15\omega + 25\omega^2 \\
 &= 12 + 35\omega + 25\omega^2 \\
 &= 12 + 35\omega + 25(-1 - \omega) \\
 &= 12 + 35\omega - 25 - 25\omega \\
 &= -13 + 10\omega
 \end{aligned}$$

Skill-building

Question 1:

We seek the cube roots of $z = 8i$. First, express $8i$ in modulus-argument form:

$$\begin{aligned}
 |z| &= 8 \\
 \arg(z) &= \frac{\pi}{2} \\
 z &= 8\text{cis}\left(\frac{\pi}{2}\right)
 \end{aligned}$$

Using the root formula $z_k = r^{1/n}\text{cis}\left(\frac{\theta+2k\pi}{n}\right)$ for $k = 0, 1, 2$:

$$\begin{aligned}
 z_0 &= 8^{1/3}\text{cis}\left(\frac{\pi/2}{3}\right) = 2\text{cis}\left(\frac{\pi}{6}\right) \\
 z_1 &= 8^{1/3}\text{cis}\left(\frac{\pi/2 + 2\pi}{3}\right) = 2\text{cis}\left(\frac{5\pi}{6}\right) \\
 z_2 &= 8^{1/3}\text{cis}\left(\frac{\pi/2 + 4\pi}{3}\right) = 2\text{cis}\left(\frac{9\pi}{6}\right) = 2\text{cis}\left(\frac{3\pi}{2}\right) = 2\text{cis}\left(-\frac{\pi}{2}\right)
 \end{aligned}$$

Question 2:

Solve $z^4 + 16 = 0 \implies z^4 = -16$.

Express -16 in modulus-argument form:

$$-16 = 16\text{cis}(\pi)$$

Using the root formula $z_k = r^{1/n}\text{cis}\left(\frac{\theta+2k\pi}{n}\right)$ for $k = 0, 1, 2, 3$:

$$\begin{aligned}
 z_0 &= 16^{1/4}\text{cis}\left(\frac{\pi}{4}\right) = 2\text{cis}\left(\frac{\pi}{4}\right) = 2\left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) = \sqrt{2} + i\sqrt{2} \\
 z_1 &= 16^{1/4}\text{cis}\left(\frac{\pi + 2\pi}{4}\right) = 2\text{cis}\left(\frac{3\pi}{4}\right) = 2\left(-\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) = -\sqrt{2} + i\sqrt{2} \\
 z_2 &= 16^{1/4}\text{cis}\left(\frac{\pi + 4\pi}{4}\right) = 2\text{cis}\left(\frac{5\pi}{4}\right) = 2\left(-\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) = -\sqrt{2} - i\sqrt{2} \\
 z_3 &= 16^{1/4}\text{cis}\left(\frac{\pi + 6\pi}{4}\right) = 2\text{cis}\left(\frac{7\pi}{4}\right) = 2\left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) = \sqrt{2} - i\sqrt{2}
 \end{aligned}$$

Plotting these on an Argand diagram, they form a square centered at the origin with vertices at $(\pm\sqrt{2}, \pm\sqrt{2})$.

Question 3a:

Given $z = \cos \theta + i \sin \theta = \text{cis}(\theta)$.

By De Moivre's Theorem:

$$\begin{aligned} z^3 &= (\text{cis}(\theta))^3 \\ &= \text{cis}(3\theta) \\ &= \cos(3\theta) + i \sin(3\theta) \end{aligned}$$

Question 3b:

Expand $(\cos \theta + i \sin \theta)^3$ using the binomial expansion $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$:

$$\begin{aligned} z^3 &= \cos^3 \theta + 3 \cos^2 \theta (i \sin \theta) + 3 \cos \theta (i \sin \theta)^2 + (i \sin \theta)^3 \\ &= \cos^3 \theta + 3i \cos^2 \theta \sin \theta + 3 \cos \theta (-\sin^2 \theta) + (-i \sin^3 \theta) \\ &= (\cos^3 \theta - 3 \cos \theta \sin^2 \theta) + i(3 \cos^2 \theta \sin \theta - \sin^3 \theta) \end{aligned}$$

Question 3c:

From parts (a) and (b), we have two expressions for z^3 :

$$\begin{aligned} \text{Real part: } \cos(3\theta) &= \cos^3 \theta - 3 \cos \theta \sin^2 \theta \\ \text{Imaginary part: } \sin(3\theta) &= 3 \cos^2 \theta \sin \theta - \sin^3 \theta \end{aligned}$$

To express $\sin(3\theta)$ solely in terms of $\sin \theta$, use $\cos^2 \theta = 1 - \sin^2 \theta$:

$$\begin{aligned} \sin(3\theta) &= 3(1 - \sin^2 \theta) \sin \theta - \sin^3 \theta \\ &= 3 \sin \theta - 3 \sin^3 \theta - \sin^3 \theta \\ &= 3 \sin \theta - 4 \sin^3 \theta \end{aligned}$$

Question 3d:

Given $\sin(3\theta) = 1$.

Solving for θ :

$$\begin{aligned} 3\theta &= \frac{\pi}{2} + 2k\pi \\ \theta &= \frac{\pi}{6} + \frac{2k\pi}{3} \end{aligned}$$

For $k = 0, 1, 2$:

$$\begin{aligned}\theta_0 &= \frac{\pi}{6} \implies x_0 = \sin\left(\frac{\pi}{6}\right) = \frac{1}{2} \\ \theta_1 &= \frac{5\pi}{6} \implies x_1 = \sin\left(\frac{5\pi}{6}\right) = \frac{1}{2} \\ \theta_2 &= \frac{3\pi}{2} \implies x_2 = \sin\left(\frac{3\pi}{2}\right) = -1\end{aligned}$$

Thus, the solutions to $-4x^3 + 3x - 1 = 0$ are $x = \frac{1}{2}$ (double root) and $x = -1$.

Question 4:

Using the real part of z^3 from Question 3(b):

$$\begin{aligned}\cos(3\theta) &= \cos^3 \theta - 3 \cos \theta \sin^2 \theta \\ &= \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta) \\ &= \cos^3 \theta - 3 \cos \theta + 3 \cos^3 \theta \\ &= 4 \cos^3 \theta - 3 \cos \theta\end{aligned}$$

Now let $\cos(3\theta) = 1$:

$$\begin{aligned}3\theta &= 0 + 2k\pi \\ \theta &= \frac{2k\pi}{3}\end{aligned}$$

For $k = 0, 1, 2$:

$$\begin{aligned}\theta_0 &= 0 \implies x_0 = \cos(0) = 1 \\ \theta_1 &= \frac{2\pi}{3} \implies x_1 = \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2} \\ \theta_2 &= \frac{4\pi}{3} \implies x_2 = \cos\left(\frac{4\pi}{3}\right) = -\frac{1}{2}\end{aligned}$$

Substituting $x = \cos \theta$ into $4x^3 - 3x - 1 = 0$, the solutions are $x = 1$ and $x = -\frac{1}{2}$ (double root).

Question 5a:

Let $z = \text{cis}(\theta)$. Then $\frac{1}{z} = z^{-1} = \text{cis}(-\theta)$.

$$\begin{aligned}z - \frac{1}{z} &= (\cos \theta + i \sin \theta) - (\cos(-\theta) + i \sin(-\theta)) \\ &= (\cos \theta + i \sin \theta) - (\cos \theta - i \sin \theta) \\ &= 2i \sin \theta\end{aligned}$$

Question 5b:

Using De Moivre's Theorem, $z^n = \text{cis}(n\theta)$ and $z^{-n} = \text{cis}(-n\theta)$.

$$\begin{aligned}
z^n - \frac{1}{z^n} &= (\cos(n\theta) + i \sin(n\theta)) - (\cos(-n\theta) + i \sin(-n\theta)) \\
&= (\cos(n\theta) + i \sin(n\theta)) - (\cos(n\theta) - i \sin(n\theta)) \\
&= 2i \sin(n\theta)
\end{aligned}$$

Question 5c:

Expand $(z - \frac{1}{z})^5$ using the binomial theorem:

$$\begin{aligned}
(z - \frac{1}{z})^5 &= z^5 - 5z^4(\frac{1}{z}) + 10z^3(\frac{1}{z})^2 - 10z^2(\frac{1}{z})^3 + 5z(\frac{1}{z})^4 - (\frac{1}{z})^5 \\
&= z^5 - 5z^3 + 10z - 10\frac{1}{z} + 5\frac{1}{z^3} - \frac{1}{z^5} \\
&= (z^5 - \frac{1}{z^5}) - 5(z^3 - \frac{1}{z^3}) + 10(z - \frac{1}{z})
\end{aligned}$$

Question 5d:

Substitute the results from part (b) into the expansion from part (c):

$$\begin{aligned}
(z - \frac{1}{z})^5 &= 2i \sin(5\theta) - 5(2i \sin(3\theta)) + 10(2i \sin \theta) \\
&= 2i \sin(5\theta) - 10i \sin(3\theta) + 20i \sin \theta
\end{aligned}$$

From part (a), $(z - \frac{1}{z})^5 = (2i \sin \theta)^5 = 32i^5 \sin^5 \theta = 32i \sin^5 \theta$ (since $i^5 = i$).
Therefore:

$$2i \sin(5\theta) - 10i \sin(3\theta) + 20i \sin \theta = 32i \sin^5 \theta$$

Question 5e:

Divide both sides by $32i$:

$$\begin{aligned}
\sin^5 \theta &= \frac{2i \sin(5\theta) - 10i \sin(3\theta) + 20i \sin \theta}{32i} \\
&= \frac{2 \sin(5\theta) - 10 \sin(3\theta) + 20 \sin \theta}{32} \\
&= \frac{1}{16}(\sin(5\theta) - 5 \sin(3\theta) + 10 \sin \theta)
\end{aligned}$$

Easier Exam Questions

Question 1:

We seek the fourth roots of $z = 2 + 2\sqrt{3}i$.

First, find the modulus and argument:

$$|z| = \sqrt{2^2 + (2\sqrt{3})^2} = \sqrt{4 + 12} = \sqrt{16} = 4$$

$$\arg(z) = \tan^{-1}\left(\frac{2\sqrt{3}}{2}\right) = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

Thus, $z = 4\text{cis}(\frac{\pi}{3})$. The fourth roots are $z_k = 4^{1/4}\text{cis}(\frac{\pi/3+2k\pi}{4})$ for $k = 0, 1, 2, 3$:

$$\begin{aligned} z_0 &= \sqrt{2}\text{cis}\left(\frac{\pi}{12}\right) \\ z_1 &= \sqrt{2}\text{cis}\left(\frac{\pi/3 + 2\pi}{4}\right) = \sqrt{2}\text{cis}\left(\frac{7\pi}{12}\right) \\ z_2 &= \sqrt{2}\text{cis}\left(\frac{\pi/3 + 4\pi}{4}\right) = \sqrt{2}\text{cis}\left(\frac{13\pi}{12}\right) = \sqrt{2}\text{cis}\left(-\frac{11\pi}{12}\right) \\ z_3 &= \sqrt{2}\text{cis}\left(\frac{\pi/3 + 6\pi}{4}\right) = \sqrt{2}\text{cis}\left(\frac{19\pi}{12}\right) = \sqrt{2}\text{cis}\left(-\frac{5\pi}{12}\right) \end{aligned}$$

Question 2:

We use the binomial expansion of $(\cos \theta + i \sin \theta)^5$:

$$\begin{aligned} (\cos \theta + i \sin \theta)^5 &= \cos^5 \theta + 5 \cos^4 \theta (i \sin \theta) + 10 \cos^3 \theta (i \sin \theta)^2 + 10 \cos^2 \theta (i \sin \theta)^3 \\ &\quad + 5 \cos \theta (i \sin \theta)^4 + (i \sin \theta)^5 \\ &= \cos^5 \theta + 5i \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta - 10i \cos^2 \theta \sin^3 \theta + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta \end{aligned}$$

Equating the real part to $\cos(5\theta)$ from De Moivre's Theorem:

$$\begin{aligned} \cos(5\theta) &= \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta \\ &= \cos^5 \theta - 10 \cos^3 \theta (1 - \cos^2 \theta) + 5 \cos \theta (1 - \cos^2 \theta)^2 \\ &= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta (1 - 2 \cos^2 \theta + \cos^4 \theta) \\ &= 11 \cos^5 \theta - 10 \cos^3 \theta + 5 \cos \theta - 10 \cos^3 \theta + 5 \cos^5 \theta \\ &= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta \end{aligned}$$

Question 3:

Given $z^5 = -a$ where $a > 0$. Express $-a$ in modulus-argument form:

$$-a = a\text{cis}(\pi)$$

One solution is $z_0 = a^{1/5}\text{cis}(\frac{\pi}{5})$. The other solutions are separated by $\frac{2\pi}{5}$:

$$\begin{aligned} z_1 &= a^{1/5}\text{cis}\left(\frac{\pi}{5} + \frac{2\pi}{5}\right) = a^{1/5}\text{cis}\left(\frac{3\pi}{5}\right) \\ z_2 &= a^{1/5}\text{cis}\left(\frac{3\pi}{5} + \frac{2\pi}{5}\right) = a^{1/5}\text{cis}\left(\frac{5\pi}{5}\right) = a^{1/5}\text{cis}(\pi) = -a^{1/5} \\ z_3 &= a^{1/5}\text{cis}\left(\frac{5\pi}{5} + \frac{2\pi}{5}\right) = a^{1/5}\text{cis}\left(\frac{7\pi}{5}\right) \\ z_4 &= a^{1/5}\text{cis}\left(\frac{7\pi}{5} + \frac{2\pi}{5}\right) = a^{1/5}\text{cis}\left(\frac{9\pi}{5}\right) \end{aligned}$$

Comparing this to the options, the real solution is $z_2 = -a^{1/5}$.

Correct Option: (B)

Question 4:

Given ω is a complex cube root of unity, so $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$.

Simplify the expression $(1 - 3\omega + \omega^2)(1 + \omega - 8\omega^2)$.

Use $1 + \omega^2 = -\omega$ in the first bracket:

$$\begin{aligned}(1 - 3\omega + \omega^2) &= (1 + \omega^2) - 3\omega \\ &= -\omega - 3\omega \\ &= -4\omega\end{aligned}$$

Use $1 + \omega = -\omega^2$ in the second bracket:

$$\begin{aligned}(1 + \omega - 8\omega^2) &= (1 + \omega) - 8\omega^2 \\ &= -\omega^2 - 8\omega^2 \\ &= -9\omega^2\end{aligned}$$

Multiply the results:

$$\begin{aligned}\text{Expr} &= (-4\omega)(-9\omega^2) \\ &= 36\omega^3 \\ &= 36(1) = 36\end{aligned}$$

Correct Option: (D)

Question 5:

Given $w^5 = 1$. We want the value of $1 + w + w^2 + w^3 + w^4 + w^5$.

This is a geometric series with $a = 1$ and $r = w$:

$$\begin{aligned}\text{Sum} &= \frac{1(w^6 - 1)}{w - 1} \\ &= \frac{w^5 \cdot w - 1}{w - 1} \\ &= \frac{1 \cdot w - 1}{w - 1} \\ &= \frac{w - 1}{w - 1} = 1\end{aligned}$$

Alternatively, since $w^5 = 1$, the sum of the first 5 roots $1 + w + w^2 + w^3 + w^4 = 0$. Thus:

$$\begin{aligned}\text{Sum} &= (1 + w + w^2 + w^3 + w^4) + w^5 \\ &= 0 + 1 = 1\end{aligned}$$

Correct Option: (A)

Harder Exam Questions

Question 1i:

Given $z = \sqrt{3} - i$.

Modulus:

$$|z| = \sqrt{(\sqrt{3})^2 + (-1)^2} = \sqrt{3+1} = 2$$

Argument:

$$\arg(z) = \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) = -\frac{\pi}{6}$$

Modulus-argument form:

$$z = 2\text{cis}\left(-\frac{\pi}{6}\right)$$

Question 1ii:

Using De Moivre's Theorem on $z = 2\text{cis}\left(-\frac{\pi}{6}\right)$:

$$\begin{aligned} z^7 &= (2\text{cis}\left(-\frac{\pi}{6}\right))^7 \\ &= 2^7 \text{cis}\left(7 \cdot -\frac{\pi}{6}\right) \\ &= 128 \text{cis}\left(-\frac{7\pi}{6}\right) \\ &= 128 \text{cis}\left(\frac{5\pi}{6}\right) \\ &= 128\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right) \\ &= 128\left(-\frac{\sqrt{3}}{2} + i\frac{1}{2}\right) \\ &= -64\sqrt{3} + 64i \end{aligned}$$

Question 1iii:

We have $z = 2\text{cis}\left(-\frac{\pi}{6}\right)$. The conjugate is $\bar{z} = 2\text{cis}\left(\frac{\pi}{6}\right)$.

Then $(\bar{z})^7 = 2^7 \text{cis}\left(\frac{7\pi}{6}\right)$.

$$\begin{aligned}
\frac{z^7}{(\bar{z})^7} &= \frac{128\text{cis}(-\frac{7\pi}{6})}{128\text{cis}(\frac{7\pi}{6})} \\
&= \text{cis}(-\frac{7\pi}{6} - \frac{7\pi}{6}) \\
&= \text{cis}(-\frac{14\pi}{6}) = \text{cis}(-\frac{7\pi}{3}) \\
&= \text{cis}(-\frac{7\pi}{3} + 2\pi) = \text{cis}(-\frac{\pi}{3}) \\
&= \cos(-\frac{\pi}{3}) + i\sin(-\frac{\pi}{3}) \\
&= \frac{1}{2} - i\frac{\sqrt{3}}{2}
\end{aligned}$$

Question 2i:

The equation is $z^5 = -1$. The roots are the 5th roots of -1 .

Express -1 as $1\text{cis}(\pi)$. The roots are $z_k = 1^{1/5}\text{cis}(\frac{\pi+2k\pi}{5})$ for $k = 0, 1, 2, 3, 4$:

$$\begin{aligned}
z_0 &= \text{cis}(\frac{\pi}{5}) \\
z_1 &= \text{cis}(\frac{3\pi}{5}) \\
z_2 &= \text{cis}(\pi) = -1 \\
z_3 &= \text{cis}(\frac{7\pi}{5}) \\
z_4 &= \text{cis}(\frac{9\pi}{5})
\end{aligned}$$

Sketch: The roots are 5 points equally spaced on the unit circle, with one point at $(-1, 0)$ and others at $\pm\frac{\pi}{5}$ and $\pm\frac{3\pi}{5}$ from the negative real axis.

Question 2ii:

We know $z^5 + 1 = (z + 1)(z^4 - z^3 + z^2 - z + 1)$.

To factor into irreducible real factors, we pair conjugate roots:

$$\text{Roots: } z = -1, \text{cis}(\frac{\pi}{5}), \text{cis}(\frac{3\pi}{5}), \text{cis}(\frac{7\pi}{5}), \text{cis}(\frac{9\pi}{5})$$

Pairing $\text{cis}(\frac{\pi}{5})$ and $\text{cis}(\frac{9\pi}{5})$:

$$\begin{aligned}
(z - \text{cis}(\frac{\pi}{5}))(z - \text{cis}(\frac{9\pi}{5})) &= z^2 - z(\text{cis}(\frac{\pi}{5}) + \text{cis}(\frac{9\pi}{5})) + 1 \\
&= z^2 - 2z\cos(\frac{\pi}{5}) + 1
\end{aligned}$$

Pairing $\text{cis}(\frac{3\pi}{5})$ and $\text{cis}(\frac{7\pi}{5})$:

$$\begin{aligned}
(z - \text{cis}(\frac{3\pi}{5}))(z - \text{cis}(\frac{7\pi}{5})) &= z^2 - z(\text{cis}(\frac{3\pi}{5}) + \text{cis}(\frac{7\pi}{5})) + 1 \\
&= z^2 - 2z\cos(\frac{3\pi}{5}) + 1
\end{aligned}$$

Thus, the factorization is:

$$z^5 + 1 = (z + 1)(z^2 - 2z \cos(\frac{\pi}{5}) + 1)(z^2 - 2z \cos(\frac{3\pi}{5}) + 1)$$

Question 2iii:

From the factorization $z^5 + 1 = (z + 1)(z^4 - z^3 + z^2 - z + 1)$, we know that the roots of $z^4 - z^3 + z^2 - z + 1 = 0$ are $\text{cis}(\frac{\pi}{5})$, $\text{cis}(\frac{3\pi}{5})$, $\text{cis}(\frac{7\pi}{5})$, $\text{cis}(\frac{9\pi}{5})$.

Sum of roots of $z^4 - z^3 + z^2 - z + 1 = 0$ is $-\frac{-1}{1} = 1$.

$$\begin{aligned} \text{Sum} &= \text{cis}(\frac{\pi}{5}) + \text{cis}(\frac{3\pi}{5}) + \text{cis}(\frac{7\pi}{5}) + \text{cis}(\frac{9\pi}{5}) \\ &= 2 \cos(\frac{\pi}{5}) + 2 \cos(\frac{3\pi}{5}) = 1 \\ \implies \cos(\frac{\pi}{5}) + \cos(\frac{3\pi}{5}) &= \frac{1}{2} \end{aligned}$$

For the product of roots, the product of all 4 roots is $\frac{1}{1} = 1$.

Pairing them as $(z^2 - 2z \cos(\frac{\pi}{5}) + 1)$ and $(z^2 - 2z \cos(\frac{3\pi}{5}) + 1)$:

$$\text{Product of constants} = (1)(1) = 1$$

$$\text{Product of } z^2 \text{ terms} = (z^2 - 2z \cos(\frac{\pi}{5}) + 1)(z^2 - 2z \cos(\frac{3\pi}{5}) + 1)$$

Coefficient of z^2

$$\text{in } z^4 - z^3 + z^2 - z + 1 \text{ is } 1.$$

Expanding the two quadratics:

$$\begin{aligned} \text{Coeff of } z^2 &= 1 + 1 + (-2 \cos(\frac{\pi}{5}))(-2 \cos(\frac{3\pi}{5})) \\ &= 2 + 4 \cos(\frac{\pi}{5}) \cos(\frac{3\pi}{5}) = 1 \\ \implies 4 \cos(\frac{\pi}{5}) \cos(\frac{3\pi}{5}) &= -1 \\ \implies \cos(\frac{\pi}{5}) \cos(\frac{3\pi}{5}) &= -\frac{1}{4} \end{aligned}$$

Question 2iv:

Let $u = \cos(\frac{\pi}{5})$ and $v = \cos(\frac{3\pi}{5})$. We have $u + v = \frac{1}{2}$ and $uv = -\frac{1}{4}$.

These are the roots of the quadratic equation $x^2 - (u + v)x + uv = 0$:

$$\begin{aligned} x^2 - \frac{1}{2}x - \frac{1}{4} &= 0 \\ \implies 2x^2 - x - \frac{1}{2} &= 0 \\ \implies 4x^2 - 2x - 1 &= 0 \end{aligned}$$

Using the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$:

$$\begin{aligned} x &= \frac{2 \pm \sqrt{(-2)^2 - 4(4)(-1)}}{2(4)} \\ &= \frac{2 \pm \sqrt{4 + 16}}{8} \\ &= \frac{2 \pm \sqrt{20}}{8} \\ &= \frac{2 \pm 2\sqrt{5}}{8} \\ &= \frac{1 \pm \sqrt{5}}{4} \end{aligned}$$

Since $\cos(\frac{\pi}{5}) > 0$ and $\cos(\frac{3\pi}{5}) < 0$:

$$\begin{aligned} \cos(\frac{\pi}{5}) &= \frac{1 + \sqrt{5}}{4} \\ \cos(\frac{3\pi}{5}) &= \frac{1 - \sqrt{5}}{4} \end{aligned}$$

Question 3:

The zeros are $\omega, \omega^3, \omega^7, \omega^9$ where $\omega = \text{cis}(\frac{\pi}{5})$.

Note that $\omega^{10} = \text{cis}(2\pi) = 1$. The zeros are $\text{cis}(\frac{\pi}{5}), \text{cis}(\frac{3\pi}{5}), \text{cis}(\frac{7\pi}{5}), \text{cis}(\frac{9\pi}{5})$.

These are exactly the roots of $z^5 + 1 = 0$ except for $z = -1$ (which is ω^5).

We know $z^5 + 1 = (z + 1)(z^4 - z^3 + z^2 - z + 1)$.

Since the zeros are the roots of $z^5 + 1 = 0$ excluding $z = -1$, the polynomial is:

$$P(z) = z^4 - z^3 + z^2 - z + 1$$

Correct Option: (D)

Question 4:

Given $z^n + \frac{1}{z^n} = 2 \cos(n\theta)$. We want to express $\cos^6 \theta$ in terms of $\cos(n\theta)$.

First, express $\cos \theta$ as $\frac{1}{2}(z + \frac{1}{z})$:

$$\begin{aligned} \cos^6 \theta &= \left[\frac{1}{2} \left(z + \frac{1}{z} \right) \right]^6 \\ &= \frac{1}{64} \left(z + \frac{1}{z} \right)^6 \end{aligned}$$

Expand $(z + \frac{1}{z})^6$ using the binomial theorem:

$$\begin{aligned} \left(z + \frac{1}{z} \right)^6 &= z^6 + 6z^5 \left(\frac{1}{z} \right) + 15z^4 \left(\frac{1}{z} \right)^2 + 20z^3 \left(\frac{1}{z} \right)^3 + 15z^2 \left(\frac{1}{z} \right)^4 + 6z \left(\frac{1}{z} \right)^5 + \frac{1}{z^6} \\ &= \left(z^6 + \frac{1}{z^6} \right) + 6 \left(z^4 + \frac{1}{z^4} \right) + 15 \left(z^2 + \frac{1}{z^2} \right) + 20 \end{aligned}$$

Substitute $z^n + \frac{1}{z^n} = 2 \cos(n\theta)$:

$$\begin{aligned}\cos^6 \theta &= \frac{1}{64} [2 \cos(6\theta) + 6(2 \cos(4\theta)) + 15(2 \cos(2\theta)) + 20] \\ &= \frac{1}{64} [2 \cos(6\theta) + 12 \cos(4\theta) + 30 \cos(2\theta) + 20] \\ &= \frac{1}{32} [\cos(6\theta) + 6 \cos(4\theta) + 15 \cos(2\theta) + 10]\end{aligned}$$

Question 5i:

Using the binomial expansion of $(\cos \theta + i \sin \theta)^5$:

$$\begin{aligned}\cos(5\theta) &= \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta \\ &= \cos^5 \theta - 10 \cos^3 \theta (1 - \cos^2 \theta) + 5 \cos \theta (1 - \cos^2 \theta)^2 \\ &= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta (1 - 2 \cos^2 \theta + \cos^4 \theta) \\ &= 11 \cos^5 \theta - 10 \cos^3 \theta + 5 \cos \theta - 10 \cos^3 \theta + 5 \cos^5 \theta \\ &= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta\end{aligned}$$

Question 5ii:

We want to solve $16x^4 - 20x^2 + 5 = 0$. Let $x = \cos \theta$.

From part (i), $\cos(5\theta) = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$.

Factor out $\cos \theta$:

$$\cos(5\theta) = \cos \theta (16 \cos^4 \theta - 20 \cos^2 \theta + 5)$$

For $16x^4 - 20x^2 + 5 = 0$, we need $\cos(5\theta) = 0$ but $\cos \theta \neq 0$.

$$\begin{aligned}5\theta &= \frac{\pi}{2} + k\pi \\ \theta &= \frac{\pi}{10} + \frac{k\pi}{5}\end{aligned}$$

For $k = 0, 1, 2, 3, 4$:

$$\begin{aligned}\theta_0 &= \frac{\pi}{10} \implies x_0 = \cos\left(\frac{\pi}{10}\right) \\ \theta_1 &= \frac{3\pi}{10} \implies x_1 = \cos\left(\frac{3\pi}{10}\right) \\ \theta_2 &= \frac{5\pi}{10} = \frac{\pi}{2} \implies \cos\left(\frac{\pi}{2}\right) = 0 \text{ (Excluded)}\end{aligned}$$

Since the equation is in x^2 , the solutions are $\pm \cos(\frac{\pi}{10})$ and $\pm \cos(\frac{3\pi}{10})$.

Question 5iii:

To find the exact value of $\cos(\frac{\pi}{10})$, we use the result from 5(ii) where $x = \cos(\frac{\pi}{10})$ is a root of $16x^4 - 20x^2 + 5 = 0$.

Let $u = x^2 = \cos^2(\frac{\pi}{10})$.

$$\begin{aligned} 16u^2 - 20u + 5 &= 0 \\ u &= \frac{20 \pm \sqrt{400 - 320}}{32} \\ &= \frac{20 \pm \sqrt{80}}{32} \\ &= \frac{20 \pm 4\sqrt{5}}{32} \\ &= \frac{5 \pm \sqrt{5}}{8} \end{aligned}$$

Since $\cos^2(\frac{\pi}{10})$ is closer to 1 than $\cos^2(\frac{3\pi}{10})$, we take the positive sign:

$$\begin{aligned} \cos^2(\frac{\pi}{10}) &= \frac{5 + \sqrt{5}}{8} \\ \cos(\frac{\pi}{10}) &= \sqrt{\frac{5 + \sqrt{5}}{8}} = \frac{\sqrt{10 + 2\sqrt{5}}}{4} \end{aligned}$$

Question 6i:

If $1, \omega_1, \omega_2$ are the cube roots of 1, then they are the roots of $z^3 - 1 = 0$.

This factors as $(z - 1)(z^2 + z + 1) = 0$.

Since ω_1, ω_2 are the non-real roots, they must satisfy $z^2 + z + 1 = 0$.

$$\begin{aligned} \omega_1^2 + \omega_1 + 1 &= 0 \\ \omega_2^2 + \omega_2 + 1 &= 0 \end{aligned}$$

Question 6ii:

To prove $x^{2n} + x^n + 1$ is divisible by $x^2 + x + 1$, we show that the roots of $x^2 + x + 1 = 0$ are also roots of $x^{2n} + x^n + 1 = 0$.

Let ω be a root of $x^2 + x + 1 = 0$. Then $\omega^2 + \omega + 1 = 0$ and $\omega^3 = 1$.

We evaluate $P(x) = x^{2n} + x^n + 1$ at $x = \omega$:

$$P(\omega) = \omega^{2n} + \omega^n + 1$$

Since n is not a multiple of 3, n can be $3k + 1$ or $3k + 2$.

Case 1: $n = 3k + 1$

$$\begin{aligned} P(\omega) &= \omega^{2(3k+1)} + \omega^{3k+1} + 1 \\ &= \omega^{6k+2} + \omega^{3k+1} + 1 \\ &= (\omega^3)^{2k} \cdot \omega^2 + (\omega^3)^k \cdot \omega + 1 \\ &= 1^{2k} \cdot \omega^2 + 1^k \cdot \omega + 1 \\ &= \omega^2 + \omega + 1 = 0 \end{aligned}$$

Case 2: $n = 3k + 2$

$$\begin{aligned}
P(\omega) &= \omega^{2(3k+2)} + \omega^{3k+2} + 1 \\
&= \omega^{6k+4} + \omega^{3k+2} + 1 \\
&= (\omega^3)^{2k+1} \cdot \omega + (\omega^3)^k \cdot \omega^2 + 1 \\
&= 1^{2k+1} \cdot \omega + 1^k \cdot \omega^2 + 1 \\
&= \omega + \omega^2 + 1 = 0
\end{aligned}$$

In both cases, $P(\omega) = 0$. Thus, $x^2 + x + 1$ is a factor of $x^{2n} + x^n + 1$.

Question 6ii:

To prove $x^{2n} + x^n + 1$ is divisible by $x^2 + x + 1$, we show that the roots of $x^2 + x + 1 = 0$ are also roots of $x^{2n} + x^n + 1 = 0$.

Let ω be a root of $x^2 + x + 1 = 0$. Then $\omega^2 + \omega + 1 = 0$ and $\omega^3 = 1$.

We evaluate $P(x) = x^{2n} + x^n + 1$ at $x = \omega$:

$$P(\omega) = \omega^{2n} + \omega^n + 1$$

Since n is not a multiple of 3, n can be $3k + 1$ or $3k + 2$.

Case 1: $n = 3k + 1$

$$\begin{aligned}
P(\omega) &= \omega^{2(3k+1)} + \omega^{3k+1} + 1 \\
&= \omega^{6k+2} + \omega^{3k+1} + 1 \\
&= (\omega^3)^{2k} \cdot \omega^2 + (\omega^3)^k \cdot \omega + 1 \\
&= 1^{2k} \cdot \omega^2 + 1^k \cdot \omega + 1 \\
&= \omega^2 + \omega + 1 = 0
\end{aligned}$$

LCase 2: $n = 3k + 2$

$$\begin{aligned}
P(\omega) &= \omega^{2(3k+2)} + \omega^{3k+2} + 1 \\
&= \omega^{6k+4} + \omega^{3k+2} + 1 \\
&= (\omega^3)^{2k+1} \cdot \omega + (\omega^3)^k \cdot \omega^2 + 1 \\
&= 1^{2k+1} \cdot \omega + 1^k \cdot \omega^2 + 1 \\
&= \omega + \omega^2 + 1 = 0
\end{aligned}$$

In both cases, $P(\omega) = 0$. Thus, $x^2 + x + 1$ is a factor of $x^{2n} + x^n + 1$.